

EX
1

Factoriser les expressions suivantes.

$$A = 3a - 6b$$

$$B = -3a + 15b$$

EX
2

Factoriser les expressions suivantes.

$$A = -10a - 35b$$

$$B = -12a + 15b$$

EX
3

Factoriser les expressions suivantes.

$$A = 44x - 55x^2$$

$$B = 55x + 66x^2$$

EX
1

Factoriser les expressions suivantes.

$$A = -11a - 33b$$

$$B = 5a + 25b$$

EX
2

Factoriser les expressions suivantes.

$$A = -18a - 21b$$

$$B = 35a + 40b$$

EX
3

Factoriser les expressions suivantes.

$$A = 3x^2 + 8x$$

$$B = 15x - 21x^2$$

EX
1

Factoriser les expressions suivantes.

$$A = -2a + 6b$$

$$B = 3a + 15b$$

EX
2

Factoriser les expressions suivantes.

$$A = -33a + 77b$$

$$B = 9a - 15b$$

EX
3

Factoriser les expressions suivantes.

$$A = -3x^2 + 8x$$

$$B = 4x^2 + x$$

EX
1

Factoriser les expressions suivantes.

$$A = 7a - 14b$$

$$B = -2a + 8b$$

EX
2

Factoriser les expressions suivantes.

$$A = 22a + 77b$$

$$B = -6a - 15b$$

EX
3

Factoriser les expressions suivantes.

$$A = -3x^2 + x$$

$$B = -4x^2 + 7x$$

EX
1

Factoriser les expressions suivantes.

$$A = -3a + 21b$$

$$B = 2a + 4b$$

EX
2

Factoriser les expressions suivantes.

$$A = -55a + 66b$$

$$B = -4a - 10b$$

EX
3

Factoriser les expressions suivantes.

$$A = -45x + 50x^2$$

$$B = 4x - 6x^2$$

EX
1

Factoriser les expressions suivantes.

$$A = -5a + 25b$$

$$B = 11a - 22b$$

EX
2

Factoriser les expressions suivantes.

$$A = -25a - 45b$$

$$B = 49a + 56b$$

EX
3

Factoriser les expressions suivantes.

$$A = 5x^2 + x$$

$$B = 3x^2 + 5x$$

EX
1

Factoriser les expressions suivantes.

$$A = -2a + 10b$$

$$B = 2a + 14b$$

EX
2

Factoriser les expressions suivantes.

$$A = 77a + 110b$$

$$B = -15a - 24b$$

EX
3

Factoriser les expressions suivantes.

$$A = 5x^2 + 6x$$

$$B = 3x^2 + x$$

EX
1

Factoriser les expressions suivantes.

$$A = 11a + 88b$$

$$B = -11a - 22b$$

EX
2

Factoriser les expressions suivantes.

$$A = 6a + 8b$$

$$B = -22a - 33b$$

EX
3

Factoriser les expressions suivantes.

$$A = 12x - 27x^2$$

$$B = -6x + 10x^2$$

EX
1

Factoriser les expressions suivantes.

$$A = 2a - 14b$$

$$B = -5a + 15b$$

EX
2

Factoriser les expressions suivantes.

$$A = -42a + 49b$$

$$B = 20a - 25b$$

EX
3

Factoriser les expressions suivantes.

$$A = -2x^2 + x$$

$$B = 15x - 18x^2$$

EX
1

Factoriser les expressions suivantes.

$$A = -11a + 77b$$

$$B = 2a - 18b$$

EX
2

Factoriser les expressions suivantes.

$$A = 22a + 33b$$

$$B = 25a - 45b$$

EX
3

Factoriser les expressions suivantes.

$$A = 2x^2 + x$$

$$B = 3x^2 + 5x$$

EX
1

Factoriser les expressions suivantes.

$$A = -2a - 14b$$

$$B = 2a + 4b$$

EX
2

Factoriser les expressions suivantes.

$$A = -99a - 110b$$

$$B = 6a + 15b$$

EX
3

Factoriser les expressions suivantes.

$$A = 2x^2 + x$$

$$B = 35x - 42x^2$$

EX
1

Factoriser les expressions suivantes.

$$A = -11a - 44b$$

$$B = 11a + 44b$$

EX
2

Factoriser les expressions suivantes.

$$A = 56a + 63b$$

$$B = -27a - 30b$$

EX
3

Factoriser les expressions suivantes.

$$A = -25x + 45x^2$$

$$B = 8x^2 + 9x$$

EX
1

Factoriser les expressions suivantes.

$$A = 7a + 21b$$

$$B = -2a - 14b$$

EX
2

Factoriser les expressions suivantes.

$$A = -21a - 24b$$

$$B = 44a + 77b$$

EX
3

Factoriser les expressions suivantes.

$$A = 21x - 70x^2$$

$$B = 4x^2 + x$$

EX
1

Factoriser les expressions suivantes.

$$A = -2a - 10b$$

$$B = 7a + 21b$$

EX
2

Factoriser les expressions suivantes.

$$A = 49a - 56b$$

$$B = -12a + 15b$$

EX
3

Factoriser les expressions suivantes.

$$A = 5x^2 + x$$

$$B = 2x^2 + 9x$$

EX
1

Factoriser les expressions suivantes.

$$A = -7a + 14b$$

$$B = 11a + 22b$$

EX
2

Factoriser les expressions suivantes.

$$A = 4a - 18b$$

$$B = 27a + 30b$$

EX
3

Factoriser les expressions suivantes.

$$A = 5x^2 + 8x$$

$$B = 24x - 27x^2$$

EX
1

Factoriser les expressions suivantes.

$$A = 3a - 6b$$

$$B = -2a - 12b$$

EX
2

Factoriser les expressions suivantes.

$$A = 4a - 10b$$

$$B = 88a + 99b$$

EX
3

Factoriser les expressions suivantes.

$$A = 7x^2 + x$$

$$B = -6x - 8x^2$$

EX
1

Factoriser les expressions suivantes.

$$A = 3a + 12b$$

$$B = -3a + 6b$$

EX
2

Factoriser les expressions suivantes.

$$A = 88a - 99b$$

$$B = 27a + 30b$$

EX
3

Factoriser les expressions suivantes.

$$A = 4x + 6x^2$$

$$B = -35x - 50x^2$$

EX
1

Factoriser les expressions suivantes.

$$A = -5a + 10b$$

$$B = 3a + 6b$$

EX
2

Factoriser les expressions suivantes.

$$A = 15a + 21b$$

$$B = -33a - 110b$$

EX
3

Factoriser les expressions suivantes.

$$A = -6x + 9x^2$$

$$B = 3x^2 + x$$

EX
1

Factoriser les expressions suivantes.

$$A = 11a + 33b$$

$$B = -7a + 28b$$

EX
2

Factoriser les expressions suivantes.

$$A = 55a + 77b$$

$$B = 6a - 16b$$

EX
3

Factoriser les expressions suivantes.

$$A = -7x^2 + 10x$$

$$B = 35x - 50x^2$$

EX
1

Factoriser les expressions suivantes.

$$A = -7a + 35b$$

$$B = 2a + 10b$$

EX
2

Factoriser les expressions suivantes.

$$A = -99a - 110b$$

$$B = -30a + 35b$$

EX
3

Factoriser les expressions suivantes.

$$A = 2x^2 + 9x$$

$$B = 8x^2 + x$$

EX
1

Factoriser les expressions suivantes.

$$A = 5a - 15b$$

$$B = -2a + 8b$$

EX
2

Factoriser les expressions suivantes.

$$A = -49a - 63b$$

$$B = 25a + 30b$$

EX
3

Factoriser les expressions suivantes.

$$A = -42x + 49x^2$$

$$B = -7x^2 + x$$

EX
1

Factoriser les expressions suivantes.

$$A = 2a + 12b$$

$$B = -11a - 22b$$

EX
2

Factoriser les expressions suivantes.

$$A = 10a - 12b$$

$$B = 33a + 44b$$

EX
3

Factoriser les expressions suivantes.

$$A = 5x^2 + x$$

$$B = 5x^2 + 8x$$

EX
1

Factoriser les expressions suivantes.

$$A = 2a + 6b$$

$$B = -7a + 14b$$

EX
2

Factoriser les expressions suivantes.

$$A = 8a - 10b$$

$$B = 10a + 14b$$

EX
3

Factoriser les expressions suivantes.

$$A = -2x^2 + x$$

$$B = 6x + 20x^2$$

EX
1

Factoriser les expressions suivantes.

$$A = -11a + 77b$$

$$B = 3a + 24b$$

EX
2

Factoriser les expressions suivantes.

$$A = -21a - 35b$$

$$B = 6a + 21b$$

EX
3

Factoriser les expressions suivantes.

$$A = 9x^2 + 10x$$

$$B = -2x^2 + x$$

EX
1

Factoriser les expressions suivantes.

$$A = -2a + 4b$$

$$B = 7a + 28b$$

EX
2

Factoriser les expressions suivantes.

$$A = 63a + 70b$$

$$B = 33a - 88b$$

EX
3

Factoriser les expressions suivantes.

$$A = 25x + 45x^2$$

$$B = 21x - 49x^2$$

EX
1

Factoriser les expressions suivantes.

$$A = 2a + 10b$$

$$B = -7a + 21b$$

EX
2

Factoriser les expressions suivantes.

$$A = 10a - 14b$$

$$B = 77a + 99b$$

EX
3

Factoriser les expressions suivantes.

$$A = 9x^2 + 10x$$

$$B = -99x + 110x^2$$

EX
1

Factoriser les expressions suivantes.

$$A = 3a - 15b$$

$$B = -3a + 6b$$

EX
2

Factoriser les expressions suivantes.

$$A = 12a - 15b$$

$$B = 22a + 33b$$

EX
3

Factoriser les expressions suivantes.

$$A = 2x^2 + x$$

$$B = 6x^2 + 7x$$

EX
1

Factoriser les expressions suivantes.

$$A = -11a + 44b$$

$$B = 5a + 40b$$

EX
2

Factoriser les expressions suivantes.

$$A = 25a + 45b$$

$$B = 6a - 16b$$

EX
3

Factoriser les expressions suivantes.

$$A = -20x - 45x^2$$

$$B = -22x + 55x^2$$

EX
1

Factoriser les expressions suivantes.

$$A = -5a - 35b$$

$$B = 7a + 14b$$

EX
2

Factoriser les expressions suivantes.

$$A = 16a + 18b$$

$$B = 6a - 10b$$

EX
3

Factoriser les expressions suivantes.

$$A = 5x^2 + x$$

$$B = -15x - 20x^2$$

EX
1

Factoriser les expressions suivantes.

$$A = 5a + 20b$$

$$B = -2a + 6b$$

EX
2

Factoriser les expressions suivantes.

$$A = -30a + 35b$$

$$B = 77a - 99b$$

EX
3

Factoriser les expressions suivantes.

$$A = 6x + 20x^2$$

$$B = 2x^2 + 7x$$



EX

1

$$\begin{aligned} 1. \quad A &= 3a - 6b \\ &= 3a - 3 \times 2b \\ &= 3(a - 2b) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= -3a + 15b \\ &= -3a + 3 \times 5b \\ &= 3(-a + 5b) \end{aligned}$$

EX

2

$$\begin{aligned} 1. \quad A &= -10a - 35b \\ &= 5 \times (-2)a - 5 \times 7b \\ &= 5(-2a - 7b) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= -12a + 15b \\ &= 3 \times (-4a) + 3 \times 5b \\ &= 3(-4a + 5b) \end{aligned}$$

EX

3

$$\begin{aligned} 1. \quad A &= 44x - 55x^2 \\ &= 11x \times 4 - 11x \times 5x \\ &= 11x(4 - 5x) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= 55x + 66x^2 \\ &= 11x \times 5 + 11x \times 6x \\ &= 11x(5 + 6x) \end{aligned}$$



EX

1

$$\begin{aligned} 1. \quad A &= -11a - 33b \\ &= -11a + (-11) \times 3b \\ &= -11(a + 3b) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= 5a + 25b \\ &= 5a + 5 \times 5b \\ &= 5(a + 5b) \end{aligned}$$

EX

2

$$\begin{aligned} 1. \quad A &= -18a - 21b \\ &= 3 \times (-6)a - 3 \times 7b \\ &= 3(-6a - 7b) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= 35a + 40b \\ &= 5 \times 7a + 5 \times 8b \\ &= 5(7a + 8b) \end{aligned}$$

EX

3

$$\begin{aligned} 1. \quad A &= 3x^2 + 8x \\ &= x \times 3x + x \times 8 \\ &= x(3x + 8) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= 15x - 21x^2 \\ &= 3x \times 5 - 3x \times 7x \\ &= 3x(5 - 7x) \end{aligned}$$



EX

1

$$\begin{aligned} 1. \quad A &= -2a + 6b \\ &= -2a + 2 \times 3b \\ &= 2(-a + 3b) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= 3a + 15b \\ &= 3a + 3 \times 5b \\ &= 3(a + 5b) \end{aligned}$$

EX

2

$$\begin{aligned} 1. \quad A &= -33a + 77b \\ &= 11 \times (-3a) + 11 \times 7b \\ &= 11(-3a + 7b) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= 9a - 15b \\ &= 3 \times 3a - 3 \times 5b \\ &= 3(3a - 5b) \end{aligned}$$

EX

3

$$\begin{aligned} 1. \quad A &= -3x^2 + 8x \\ &= x \times (-3)x + x \times 8 \\ &= x(-3x + 8) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= 4x^2 + x \\ &= x \times 4x + x \times 1 \\ &= x(4x + 1) \end{aligned}$$



EX

1

$$\begin{aligned} 1. \quad A &= 7a - 14b \\ &= 7a - 7 \times 2b \\ &= 7(a - 2b) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= -2a + 8b \\ &= -2a + 2 \times 4b \\ &= 2(-a + 4b) \end{aligned}$$

EX

2

$$\begin{aligned} 1. \quad A &= 22a + 77b \\ &= 11 \times 2a + 11 \times 7b \\ &= 11(2a + 7b) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= -6a - 15b \\ &= 3 \times (-2)a - 3 \times 5b \\ &= 3(-2a - 5b) \end{aligned}$$

EX

3

$$\begin{aligned} 1. \quad A &= -3x^2 + x \\ &= x \times (-3)x + x \times 1 \\ &= x(-3x + 1) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= -4x^2 + 7x \\ &= x \times (-4)x + x \times 7 \\ &= x(-4x + 7) \end{aligned}$$



EX

1

$$\begin{aligned} 1. \quad A &= -3a + 21b \\ &= -3a + 3 \times 7b \\ &= 3(-a + 7b) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= 2a + 4b \\ &= 2a + 2 \times 2b \\ &= 2(a + 2b) \end{aligned}$$

EX

2

$$\begin{aligned} 1. \quad A &= -55a + 66b \\ &= 11 \times (-5a) + 11 \times 6b \\ &= 11(-5a + 6b) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= -4a - 10b \\ &= 2 \times (-2)a - 2 \times 5b \\ &= 2(-2a - 5b) \end{aligned}$$

EX

3

$$\begin{aligned} 1. \quad A &= -45x + 50x^2 \\ &= 5x \times (-9) + 5x \times 10x \\ &= 5x(-9 + 10x) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= 4x - 6x^2 \\ &= 2x \times 2 - 2x \times 3x \\ &= 2x(2 - 3x) \end{aligned}$$



EX

1

$$\begin{aligned} 1. \quad A &= -5a + 25b \\ &= -5a + 5 \times 5b \\ &= 5(-a + 5b) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= 11a - 22b \\ &= 11a - 11 \times 2b \\ &= 11(a - 2b) \end{aligned}$$

EX

2

$$\begin{aligned} 1. \quad A &= -25a - 45b \\ &= 5 \times (-5)a - 5 \times 9b \\ &= 5(-5a - 9b) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= 49a + 56b \\ &= 7 \times 7a + 7 \times 8b \\ &= 7(7a + 8b) \end{aligned}$$

EX

3

$$\begin{aligned} 1. \quad A &= 5x^2 + x \\ &= x \times 5x + x \times 1 \\ &= x(5x + 1) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= 3x^2 + 5x \\ &= x \times 3x + x \times 5 \\ &= x(3x + 5) \end{aligned}$$



EX

1

$$\begin{aligned} 1. \quad A &= -2a + 10b \\ &= -2a + 2 \times 5b \\ &= 2(-a + 5b) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= 2a + 14b \\ &= 2a + 2 \times 7b \\ &= 2(a + 7b) \end{aligned}$$

EX

2

$$\begin{aligned} 1. \quad A &= 77a + 110b \\ &= 11 \times 7a + 11 \times 10b \\ &= 11(7a + 10b) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= -15a - 24b \\ &= 3 \times (-5)a - 3 \times 8b \\ &= 3(-5a - 8b) \end{aligned}$$

EX

3

$$\begin{aligned} 1. \quad A &= 5x^2 + 6x \\ &= x \times 5x + x \times 6 \\ &= x(5x + 6) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= 3x^2 + x \\ &= x \times 3x + x \times 1 \\ &= x(3x + 1) \end{aligned}$$



EX

1

$$\begin{aligned} 1. \quad A &= 11a + 88b \\ &= 11a + 11 \times 8b \\ &= 11(a + 8b) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= -11a - 22b \\ &= -11a + (-11) \times 2b \\ &= -11(a + 2b) \end{aligned}$$

EX

2

$$\begin{aligned} 1. \quad A &= 6a + 8b \\ &= 2 \times 3a + 2 \times 4b \\ &= 2(3a + 4b) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= -22a - 33b \\ &= 11 \times (-2)a - 11 \times 3b \\ &= 11(-2a - 3b) \end{aligned}$$

EX

3

$$\begin{aligned} 1. \quad A &= 12x - 27x^2 \\ &= 3x \times 4 - 3x \times 9x \\ &= 3x(4 - 9x) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= -6x + 10x^2 \\ &= 2x \times (-3) + 2x \times 5x \\ &= 2x(-3 + 5x) \end{aligned}$$



EX

1

$$\begin{aligned} 1. \quad A &= 2a - 14b \\ &= 2a - 2 \times 7b \\ &= 2(a - 7b) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= -5a + 15b \\ &= -5a + 5 \times 3b \\ &= 5(-a + 3b) \end{aligned}$$

EX

2

$$\begin{aligned} 1. \quad A &= -42a + 49b \\ &= 7 \times (-6a) + 7 \times 7b \\ &= 7(-6a + 7b) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= 20a - 25b \\ &= 5 \times 4a - 5 \times 5b \\ &= 5(4a - 5b) \end{aligned}$$

EX

3

$$\begin{aligned} 1. \quad A &= -2x^2 + x \\ &= x \times (-2)x + x \times 1 \\ &= x(-2x + 1) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= 15x - 18x^2 \\ &= 3x \times 5 - 3x \times 6x \\ &= 3x(5 - 6x) \end{aligned}$$



EX

1

$$\begin{aligned} 1. \quad A &= -11a + 77b \\ &= -11a + 11 \times 7b \\ &= 11(-a + 7b) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= 2a - 18b \\ &= 2a - 2 \times 9b \\ &= 2(a - 9b) \end{aligned}$$

EX

2

$$\begin{aligned} 1. \quad A &= 22a + 33b \\ &= 11 \times 2a + 11 \times 3b \\ &= 11(2a + 3b) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= 25a - 45b \\ &= 5 \times 5a - 5 \times 9b \\ &= 5(5a - 9b) \end{aligned}$$

EX

3

$$\begin{aligned} 1. \quad A &= 2x^2 + x \\ &= x \times 2x + x \times 1 \\ &= x(2x + 1) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= 3x^2 + 5x \\ &= x \times 3x + x \times 5 \\ &= x(3x + 5) \end{aligned}$$



EX

1

$$\begin{aligned} 1. \quad A &= -2a - 14b \\ &= -2a + (-2) \times 7b \\ &= -2(a + 7b) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= 2a + 4b \\ &= 2a + 2 \times 2b \\ &= 2(a + 2b) \end{aligned}$$

EX

2

$$\begin{aligned} 1. \quad A &= -99a - 110b \\ &= 11 \times (-9)a - 11 \times 10b \\ &= 11(-9a - 10b) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= 6a + 15b \\ &= 3 \times 2a + 3 \times 5b \\ &= 3(2a + 5b) \end{aligned}$$

EX

3

$$\begin{aligned} 1. \quad A &= 2x^2 + x \\ &= x \times 2x + x \times 1 \\ &= x(2x + 1) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= 35x - 42x^2 \\ &= 7x \times 5 - 7x \times 6x \\ &= 7x(5 - 6x) \end{aligned}$$



EX

1

$$\begin{aligned} 1. \quad A &= -11a - 44b \\ &= -11a + (-11) \times 4b \\ &= -11(a + 4b) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= 11a + 44b \\ &= 11a + 11 \times 4b \\ &= 11(a + 4b) \end{aligned}$$

EX

2

$$\begin{aligned} 1. \quad A &= 56a + 63b \\ &= 7 \times 8a + 7 \times 9b \\ &= 7(8a + 9b) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= -27a - 30b \\ &= 3 \times (-9)a - 3 \times 10b \\ &= 3(-9a - 10b) \end{aligned}$$

EX

3

$$\begin{aligned} 1. \quad A &= -25x + 45x^2 \\ &= 5x \times (-5) + 5x \times 9x \\ &= 5x(-5 + 9x) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= 8x^2 + 9x \\ &= x \times 8x + x \times 9 \\ &= x(8x + 9) \end{aligned}$$



EX

1

$$\begin{aligned} 1. \quad A &= 7a + 21b \\ &= 7a + 7 \times 3b \\ &= 7(a + 3b) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= -2a - 14b \\ &= -2a + (-2) \times 7b \\ &= -2(a + 7b) \end{aligned}$$

EX

2

$$\begin{aligned} 1. \quad A &= -21a - 24b \\ &= 3 \times (-7)a - 3 \times 8b \\ &= 3(-7a - 8b) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= 44a + 77b \\ &= 11 \times 4a + 11 \times 7b \\ &= 11(4a + 7b) \end{aligned}$$

EX

3

$$\begin{aligned} 1. \quad A &= 21x - 70x^2 \\ &= 7x \times 3 - 7x \times 10x \\ &= 7x(3 - 10x) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= 4x^2 + x \\ &= x \times 4x + x \times 1 \\ &= x(4x + 1) \end{aligned}$$



EX

1

$$\begin{aligned} 1. \quad A &= -2a - 10b \\ &= -2a + (-2) \times 5b \\ &= -2(a + 5b) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= 7a + 21b \\ &= 7a + 7 \times 3b \\ &= 7(a + 3b) \end{aligned}$$

EX

2

$$\begin{aligned} 1. \quad A &= 49a - 56b \\ &= 7 \times 7a - 7 \times 8b \\ &= 7(7a - 8b) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= -12a + 15b \\ &= 3 \times (-4a) + 3 \times 5b \\ &= 3(-4a + 5b) \end{aligned}$$

EX

3

$$\begin{aligned} 1. \quad A &= 5x^2 + x \\ &= x \times 5x + x \times 1 \\ &= x(5x + 1) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= 2x^2 + 9x \\ &= x \times 2x + x \times 9 \\ &= x(2x + 9) \end{aligned}$$



EX

1

$$\begin{aligned} 1. \quad A &= -7a + 14b \\ &= -7a + 7 \times 2b \\ &= 7(-a + 2b) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= 11a + 22b \\ &= 11a + 11 \times 2b \\ &= 11(a + 2b) \end{aligned}$$

EX

2

$$\begin{aligned} 1. \quad A &= 4a - 18b \\ &= 2 \times 2a - 2 \times 9b \\ &= 2(2a - 9b) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= 27a + 30b \\ &= 3 \times 9a + 3 \times 10b \\ &= 3(9a + 10b) \end{aligned}$$

EX

3

$$\begin{aligned} 1. \quad A &= 5x^2 + 8x \\ &= x \times 5x + x \times 8 \\ &= x(5x + 8) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= 24x - 27x^2 \\ &= 3x \times 8 - 3x \times 9x \\ &= 3x(8 - 9x) \end{aligned}$$



EX

1

$$\begin{aligned} 1. \quad A &= 3a - 6b \\ &= 3a - 3 \times 2b \\ &= 3(a - 2b) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= -2a - 12b \\ &= -2a + (-2) \times 6b \\ &= -2(a + 6b) \end{aligned}$$

EX

2

$$\begin{aligned} 1. \quad A &= 4a - 10b \\ &= 2 \times 2a - 2 \times 5b \\ &= 2(2a - 5b) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= 88a + 99b \\ &= 11 \times 8a + 11 \times 9b \\ &= 11(8a + 9b) \end{aligned}$$

EX

3

$$\begin{aligned} 1. \quad A &= 7x^2 + x \\ &= x \times 7x + x \times 1 \\ &= x(7x + 1) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= -6x - 8x^2 \\ &= 2x \times (-3) - 2x \times 4x \\ &= 2x(-3 - 4x) \end{aligned}$$



EX

1

$$\begin{aligned} 1. \quad A &= 3a + 12b \\ &= 3a + 3 \times 4b \\ &= 3(a + 4b) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= -3a + 6b \\ &= -3a + 3 \times 2b \\ &= 3(-a + 2b) \end{aligned}$$

EX

2

$$\begin{aligned} 1. \quad A &= 88a - 99b \\ &= 11 \times 8a - 11 \times 9b \\ &= 11(8a - 9b) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= 27a + 30b \\ &= 3 \times 9a + 3 \times 10b \\ &= 3(9a + 10b) \end{aligned}$$

EX

3

$$\begin{aligned} 1. \quad A &= 4x + 6x^2 \\ &= 2x \times 2 + 2x \times 3x \\ &= 2x(2 + 3x) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= -35x - 50x^2 \\ &= 5x \times (-7) - 5x \times 10x \\ &= 5x(-7 - 10x) \end{aligned}$$



EX

1

$$\begin{aligned} 1. \quad A &= -5a + 10b \\ &= -5a + 5 \times 2b \\ &= 5(-a + 2b) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= 3a + 6b \\ &= 3a + 3 \times 2b \\ &= 3(a + 2b) \end{aligned}$$

EX

2

$$\begin{aligned} 1. \quad A &= 15a + 21b \\ &= 3 \times 5a + 3 \times 7b \\ &= 3(5a + 7b) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= -33a - 110b \\ &= 11 \times (-3)a - 11 \times 10b \\ &= 11(-3a - 10b) \end{aligned}$$

EX

3

$$\begin{aligned} 1. \quad A &= -6x + 9x^2 \\ &= 3x \times (-2) + 3x \times 3x \\ &= 3x(-2 + 3x) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= 3x^2 + x \\ &= x \times 3x + x \times 1 \\ &= x(3x + 1) \end{aligned}$$



EX

1

$$\begin{aligned} 1. \quad A &= 11a + 33b \\ &= 11a + 11 \times 3b \\ &= 11(a + 3b) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= -7a + 28b \\ &= -7a + 7 \times 4b \\ &= 7(-a + 4b) \end{aligned}$$

EX

2

$$\begin{aligned} 1. \quad A &= 55a + 77b \\ &= 11 \times 5a + 11 \times 7b \\ &= 11(5a + 7b) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= 6a - 16b \\ &= 2 \times 3a - 2 \times 8b \\ &= 2(3a - 8b) \end{aligned}$$

EX

3

$$\begin{aligned} 1. \quad A &= -7x^2 + 10x \\ &= x \times (-7)x + x \times 10 \\ &= x(-7x + 10) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= 35x - 50x^2 \\ &= 5x \times 7 - 5x \times 10x \\ &= 5x(7 - 10x) \end{aligned}$$



EX

1

$$\begin{aligned} 1. \quad A &= -7a + 35b \\ &= -7a + 7 \times 5b \\ &= 7(-a + 5b) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= 2a + 10b \\ &= 2a + 2 \times 5b \\ &= 2(a + 5b) \end{aligned}$$

EX

2

$$\begin{aligned} 1. \quad A &= -99a - 110b \\ &= 11 \times (-9)a - 11 \times 10b \\ &= 11(-9a - 10b) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= -30a + 35b \\ &= 5 \times (-6a) + 5 \times 7b \\ &= 5(-6a + 7b) \end{aligned}$$

EX

3

$$\begin{aligned} 1. \quad A &= 2x^2 + 9x \\ &= x \times 2x + x \times 9 \\ &= x(2x + 9) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= 8x^2 + x \\ &= x \times 8x + x \times 1 \\ &= x(8x + 1) \end{aligned}$$



EX

1

$$\begin{aligned} 1. \quad A &= 5a - 15b \\ &= 5a - 5 \times 3b \\ &= 5(a - 3b) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= -2a + 8b \\ &= -2a + 2 \times 4b \\ &= 2(-a + 4b) \end{aligned}$$

EX

2

$$\begin{aligned} 1. \quad A &= -49a - 63b \\ &= 7 \times (-7)a - 7 \times 9b \\ &= 7(-7a - 9b) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= 25a + 30b \\ &= 5 \times 5a + 5 \times 6b \\ &= 5(5a + 6b) \end{aligned}$$

EX

3

$$\begin{aligned} 1. \quad A &= -42x + 49x^2 \\ &= 7x \times (-6) + 7x \times 7x \\ &= 7x(-6 + 7x) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= -7x^2 + x \\ &= x \times (-7)x + x \times 1 \\ &= x(-7x + 1) \end{aligned}$$



EX

1

$$\begin{aligned} 1. \quad A &= 2a + 12b \\ &= 2a + 2 \times 6b \\ &= 2(a + 6b) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= -11a - 22b \\ &= -11a + (-11) \times 2b \\ &= -11(a + 2b) \end{aligned}$$

EX

2

$$\begin{aligned} 1. \quad A &= 10a - 12b \\ &= 2 \times 5a - 2 \times 6b \\ &= 2(5a - 6b) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= 33a + 44b \\ &= 11 \times 3a + 11 \times 4b \\ &= 11(3a + 4b) \end{aligned}$$

EX

3

$$\begin{aligned} 1. \quad A &= 5x^2 + x \\ &= x \times 5x + x \times 1 \\ &= x(5x + 1) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= 5x^2 + 8x \\ &= x \times 5x + x \times 8 \\ &= x(5x + 8) \end{aligned}$$



EX

1

$$\begin{aligned} 1. \quad A &= 2a + 6b \\ &= 2a + 2 \times 3b \\ &= 2(a + 3b) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= -7a + 14b \\ &= -7a + 7 \times 2b \\ &= 7(-a + 2b) \end{aligned}$$

EX

2

$$\begin{aligned} 1. \quad A &= 8a - 10b \\ &= 2 \times 4a - 2 \times 5b \\ &= 2(4a - 5b) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= 10a + 14b \\ &= 2 \times 5a + 2 \times 7b \\ &= 2(5a + 7b) \end{aligned}$$

EX

3

$$\begin{aligned} 1. \quad A &= -2x^2 + x \\ &= x \times (-2)x + x \times 1 \\ &= x(-2x + 1) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= 6x + 20x^2 \\ &= 2x \times 3 + 2x \times 10x \\ &= 2x(3 + 10x) \end{aligned}$$



EX

1

$$\begin{aligned} 1. \quad A &= -11a + 77b \\ &= -11a + 11 \times 7b \\ &= 11(-a + 7b) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= 3a + 24b \\ &= 3a + 3 \times 8b \\ &= 3(a + 8b) \end{aligned}$$

EX

2

$$\begin{aligned} 1. \quad A &= -21a - 35b \\ &= 7 \times (-3)a - 7 \times 5b \\ &= 7(-3a - 5b) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= 6a + 21b \\ &= 3 \times 2a + 3 \times 7b \\ &= 3(2a + 7b) \end{aligned}$$

EX

3

$$\begin{aligned} 1. \quad A &= 9x^2 + 10x \\ &= x \times 9x + x \times 10 \\ &= x(9x + 10) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= -2x^2 + x \\ &= x \times (-2)x + x \times 1 \\ &= x(-2x + 1) \end{aligned}$$



EX

1

$$\begin{aligned} 1. \quad A &= -2a + 4b \\ &= -2a + 2 \times 2b \\ &= 2(-a + 2b) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= 7a + 28b \\ &= 7a + 7 \times 4b \\ &= 7(a + 4b) \end{aligned}$$

EX

2

$$\begin{aligned} 1. \quad A &= 63a + 70b \\ &= 7 \times 9a + 7 \times 10b \\ &= 7(9a + 10b) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= 33a - 88b \\ &= 11 \times 3a - 11 \times 8b \\ &= 11(3a - 8b) \end{aligned}$$

EX

3

$$\begin{aligned} 1. \quad A &= 25x + 45x^2 \\ &= 5x \times 5 + 5x \times 9x \\ &= 5x(5 + 9x) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= 21x - 49x^2 \\ &= 7x \times 3 - 7x \times 7x \\ &= 7x(3 - 7x) \end{aligned}$$



EX

1

$$\begin{aligned} 1. \quad A &= 2a + 10b \\ &= 2a + 2 \times 5b \\ &= 2(a + 5b) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= -7a + 21b \\ &= -7a + 7 \times 3b \\ &= 7(-a + 3b) \end{aligned}$$

EX

2

$$\begin{aligned} 1. \quad A &= 10a - 14b \\ &= 2 \times 5a - 2 \times 7b \\ &= 2(5a - 7b) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= 77a + 99b \\ &= 11 \times 7a + 11 \times 9b \\ &= 11(7a + 9b) \end{aligned}$$

EX

3

$$\begin{aligned} 1. \quad A &= 9x^2 + 10x \\ &= x \times 9x + x \times 10 \\ &= x(9x + 10) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= -99x + 110x^2 \\ &= 11x \times (-9) + 11x \times 10x \\ &= 11x(-9 + 10x) \end{aligned}$$



EX

1

$$\begin{aligned} 1. \quad A &= 3a - 15b \\ &= 3a - 3 \times 5b \\ &= 3(a - 5b) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= -3a + 6b \\ &= -3a + 3 \times 2b \\ &= 3(-a + 2b) \end{aligned}$$

EX

2

$$\begin{aligned} 1. \quad A &= 12a - 15b \\ &= 3 \times 4a - 3 \times 5b \\ &= 3(4a - 5b) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= 22a + 33b \\ &= 11 \times 2a + 11 \times 3b \\ &= 11(2a + 3b) \end{aligned}$$

EX

3

$$\begin{aligned} 1. \quad A &= 2x^2 + x \\ &= x \times 2x + x \times 1 \\ &= x(2x + 1) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= 6x^2 + 7x \\ &= x \times 6x + x \times 7 \\ &= x(6x + 7) \end{aligned}$$



EX

1

$$\begin{aligned} 1. \quad A &= -11a + 44b \\ &= -11a + 11 \times 4b \\ &= 11(-a + 4b) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= 5a + 40b \\ &= 5a + 5 \times 8b \\ &= 5(a + 8b) \end{aligned}$$

EX

2

$$\begin{aligned} 1. \quad A &= 25a + 45b \\ &= 5 \times 5a + 5 \times 9b \\ &= 5(5a + 9b) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= 6a - 16b \\ &= 2 \times 3a - 2 \times 8b \\ &= 2(3a - 8b) \end{aligned}$$

EX

3

$$\begin{aligned} 1. \quad A &= -20x - 45x^2 \\ &= 5x \times (-4) - 5x \times 9x \\ &= 5x(-4 - 9x) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= -22x + 55x^2 \\ &= 11x \times (-2) + 11x \times 5x \\ &= 11x(-2 + 5x) \end{aligned}$$

EX
1

$$\begin{aligned} 1. \quad A &= -5a - 35b \\ &= -5a + (-5) \times 7b \\ &= -5(a + 7b) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= 7a + 14b \\ &= 7a + 7 \times 2b \\ &= 7(a + 2b) \end{aligned}$$

EX
2

$$\begin{aligned} 1. \quad A &= 16a + 18b \\ &= 2 \times 8a + 2 \times 9b \\ &= 2(8a + 9b) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= 6a - 10b \\ &= 2 \times 3a - 2 \times 5b \\ &= 2(3a - 5b) \end{aligned}$$

EX
3

$$\begin{aligned} 1. \quad A &= 5x^2 + x \\ &= x \times 5x + x \times 1 \\ &= x(5x + 1) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= -15x - 20x^2 \\ &= 5x \times (-3) - 5x \times 4x \\ &= 5x(-3 - 4x) \end{aligned}$$



EX

1

$$\begin{aligned} 1. \quad A &= 5a + 20b \\ &= 5a + 5 \times 4b \\ &= 5(a + 4b) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= -2a + 6b \\ &= -2a + 2 \times 3b \\ &= 2(-a + 3b) \end{aligned}$$

EX

2

$$\begin{aligned} 1. \quad A &= -30a + 35b \\ &= 5 \times (-6a) + 5 \times 7b \\ &= 5(-6a + 7b) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= 77a - 99b \\ &= 11 \times 7a - 11 \times 9b \\ &= 11(7a - 9b) \end{aligned}$$

EX

3

$$\begin{aligned} 1. \quad A &= 6x + 20x^2 \\ &= 2x \times 3 + 2x \times 10x \\ &= 2x(3 + 10x) \end{aligned}$$

$$\begin{aligned} 2. \quad B &= 2x^2 + 7x \\ &= x \times 2x + x \times 7 \\ &= x(2x + 7) \end{aligned}$$